UNIVERSITY OF SZEGED
FACULTY OF AGRICULTURE

Engineering Internship Report

Plant Production and Horticulture Practice
Greenhouse Tomato Production

Student: *[Full Name]*

Course: Agricultural Engineering BSc

Internship place: Szentesi Paradicsom Kft., Fűz-telep, Szegvár, Hungary

Internship period: *[start date – end date (6-week practice period)]*

Internship leader: *[Name and professional status/title – e.g. Ákos Magyar, Greenhouse Production Manager, or Dr. Katona Magdolna, Managing Director. Use whoever directly supervised you on-site.]*

Hódmezővásárhely
[Year]

Contents

1 Circumstances of the Establishment of the Company

Szentesi Paradicsom Kft. was officially registered on 9 July 2013. Rather than being founded by a single entrepreneur, the company was established jointly by 27 private investors. Dr. Katona Magdolna, the current managing director, has served as co-owner and executive manager since the company's founding [1].

The company belongs to the Árpád corporate group, alongside Árpád-Agrár Zrt. and Árpád-Masterplant Kft. The group's vegetable sales are channeled through the Szentes-based Délkertész producer organization, which Árpád-Agrár Zrt. helped establish and remains its largest supplier to. Legally, Szentesi Paradicsom Kft. is a Kft. (Korlátolt Felelősségű Társaság – a private limited liability company), not a cooperative or a publicly traded entity; it operates as a professionally managed agribusiness within a larger corporate group rather than as an independent family farm.

The company was founded specifically to pursue forced/greenhouse tomato cultivation, taking advantage of favourable financing conditions available at the time, notably low-interest credit through Hungary's Növekedési Hitelprogram (*Funding for Growth Scheme*), administered by the National Bank of Hungary. The founders adopted a 15-year strategic development plan carried out in stages, with a SWOT analysis conducted after each phase to guide subsequent investment decisions – indicating a deliberately planned, long-horizon investment rather than an organic small-farm start-up.

Tomatoes are the company's sole product line: large-truss (cluster) tomatoes and cocktail/plum tomatoes, grown hydroponically in substrate (soilless culture), with nutrients delivered via fertigation and heat supplied by geothermal water. The Szegvár site houses one of Hungary's largest high-tech greenhouse complexes.

2 Organizational Structure of the Company

The Managing Director is Dr. Katona Magdolna. The company is organized into four operational divisions, each headed by its own division manager and supported by a dedicated technician. The greenhouse production manager is Ákos Magyar. At crop level, the operation is further split into three departments: young plants (nursery), tomatoes, and paprika.

[Confirm with the company whether the "four divisions" and "three departments" refer to the same organizational breakdown viewed from two angles (e.g. operational/geographic divisions vs. crop-based departments) or to genuinely separate structures, and obtain the full list of division managers to build a proper organogram. Insert the organogram as a figure in the Appendix (uncomment the example in appendix.tex) and reference it here, e.g. `Figure` **??***, once it exists.]*

2.1 Research Institute Activity

Not applicable – the internship took place at a commercial greenhouse operation, not a research institute or experimental station.

3 Human Resource Characteristics of the Company

Staffing in the tomato greenhouse is seasonal, with headcount ranging from roughly 30 to 40 workers depending on the time of year. Approximately 10–15 seasonal workers supplement the permanent team during peak periods. No formal horticultural qualifications are required of greenhouse workers; instead, staff receive hands-on training in plant care tasks during onboarding. Beyond this initial training, ongoing staff development focuses on working speed and proficiency rather than further formal instruction.

4 Founding Capital, Fixed and Current Assets

The company's fixed assets consist of five greenhouses totalling approximately 14.5 hectares company-wide, together with tractors, machinery, and associated equipment. *[The precise booking value of founding capital and current assets (inventories, receivables) was not disclosed in the interview and should be requested from the company's finance department if needed for the essay.]*

5 Liabilities and Financing

The company has not relied on commercial bank loans to finance greenhouse construction. Instead, it has drawn on government/EU support (see Section 19, Use of Subsidy) to fund new construction, equipment, and input materials. Historically, the company's founding itself was financed through low-interest credit under Hungary's Növekedési Hitelprogram, obtained at a time when those conditions were particularly favourable for launching greenhouse tomato production.

6 General Description of the Plant Production Unit

The Fűz-telep site in Szegvár is one of Hungary's largest high-tech greenhouse complexes. Company-wide, the operation spans five greenhouses totalling approximately 14.5 hectares, covering nursery (young plant), tomato, and paprika production. Of this, 4.1 hectares at the Szegvár site are dedicated specifically to tomato production, which is the focus of this report. The greenhouses are Venlo-style glasshouses, approximately 89% glass-covered, heated and cooled to maintain controlled growing conditions year-round.

7 Ecological Conditions of Production

- **Climatic conditions:** Winters in the Szegvár region are relatively mild but can reach as low as −20°C; summers are hot and dry, reaching up to 40°C. The greenhouse climate is fully controlled independently of these extremes (see below).
- **Production area:** 4.1 hectares under tomato production.

- **Soil properties:** Not applicable – production takes place in substrate, not in open soil.
- **Growing media properties:** Coir (coconut fibre), coco peat, and coir dust, chemically buffered with calcium nitrate before use (see Section 12).

Heating is supplied by geothermal water drawn from a well approximately 2,000 metres deep, typical of the Szentes–Szegvár region, which is nationally known for its geothermal resources; the water arrives at approximately 80°C. Heat is distributed through three separate pipe circuits: ground heating pipes (around 35°C, mainly used during the autumn and spring transitional seasons), grow-pipe heating running through the crop canopy (maximum 45°C, typically operated around 30°C), and top/rail heating pipes (maximum 60°C, typically operated around 50°C). Ventilation is automated: roof vents open according to canopy density to keep relative humidity within a 60–70% target range. Deviations from this range have specific consequences for the crop: excess humidity promotes grey mould (*Botrytis*); excessive heat can cause flower abortion; excessive cold slows growth and hardens plant tissue; and insufficient light results in weak plant growth.

8 Analysis of Production Activity and Products

Tomatoes are the company's sole product line, grown hydroponically in substrate with fertigation and geothermal heating. Two product types are grown: large-truss (cluster) tomatoes and cocktail/plum tomatoes. Fruit quality is evaluated on the basis of colour (red), shape (round), and firmness (solid), and this standard is applied consistently from grading through to the retail quality specification demanded by buyers (Section 17).

9 Greenhouse Types

The production greenhouses are Venlo-style glasshouses, approximately 89% glass-covered, combining high light transmission with the structural heating/cooling and ventilation systems described in Section 7. The production season runs from planting around 20 January through to the end of November.

10 Description of Production Profile

Tomatoes are the sole crop of this production unit, split between two product lines: large-truss (cluster) tomatoes and cocktail/plum tomatoes. Production is scheduled as a single extended annual cycle, from planting in mid/late January through harvest to the end of November, after which the greenhouse is cleared and disinfected ahead of the next cycle (Section 12).

11 Cultivated Varieties and Biological Bases

The company grows two main tomato types, both selected primarily for their high yield potential:

- **Large-truss (cluster) tomatoes** – reported yield of approximately 65 kg/m^2.
- **Cocktail/plum tomatoes** – the 'Ardiles Reina' variety from Dutch breeder Enza Zaden, yielding approximately 40–45 kg/m^2.

Both figures refer to the 4.1-hectare tomato block at the Szegvár site. Seeds and seedlings are sourced from the Netherlands. The company also evaluates new rootstocks on-site, including DR0141TX, a high-vigour rootstock developed by De Ruiter (Bayer) for high-tech, artificially lit glasshouse tomato production (see Section 14, Introduction of Experiments).

12 Cultivation Technology

12.1 Substrate Cultivation

Raw, unwashed coir naturally has a high cation exchange capacity and arrives saturated with sodium and potassium ions. Without treatment, the coir fibres would bind calcium and magnesium from the standard fertilizer solution – creating deficiencies of those nutrients while releasing excess sodium into the root zone. To prevent this, calcium nitrate is added as a buffering step before planting: it does not supply potassium or iron, but instead displaces the excess sodium and potassium bound to the substrate so they can be flushed out, while pre-loading the coir with calcium.

12.2 Greenhouse Preparation

At the end of each production cycle (late November), the greenhouse is fully cleared and disinfected using a sodium hypochlorite solution before substrate buffering and the next planting cycle begin.

12.3 Nutrient Supply

Fertigation is fully automated, using mixing tanks that feed a central dosing system integrated with the Priva climate and irrigation control platform (Section 13).

12.4 Planting Parameters

Planting begins around 20 January. Planting density differs by product line: large-truss tomatoes are planted at 3.7 stems/m^2, and cocktail tomatoes at 4.2 stems/m^2.

12.5 Plant Care

Ongoing plant care consists of pruning, de-leafing, stem twisting, and clipping. Pollination is carried out naturally using bumblebee colonies, the standard method in enclosed greenhouse tomato production. Irrigation is applied at approximately 10 L/m^2 per day, equivalent to roughly 400 m^3 per day across the production area during peak summer demand.

12.6 Plant Protection

The following pests and diseases have been reported on-site: whitefly (glasshouse whitefly), *Helicoverpa armigera* (cotton bollworm), aphids, tomato brown rugose fruit virus (ToBRFV), a mosaic-type virus reported as "fruit mosaic virus" *[likely Tomato mosaic virus (ToMV) – to be confirmed with the company]*, Pepino mosaic virus (PMV), cucumber mosaic virus (CMV), bacterial canker (*Clavibacter michiganensis* subsp. *michiganensis*), and *Tuta absoluta* (tomato leafminer).

Pest and disease control relies primarily on biological control, supported by yellow sticky traps for monitoring. Specific biological agents in use include *Aphidius colemani* (a parasitic wasp) against aphids, and *Encarsia formosa* (a parasitic wasp) against glasshouse whitefly.

12.7 Harvesting and Storage

Harvest onset depends on variety and is measured from first flowering: approximately 80–90 days for large-truss tomatoes and 60–65 days for cocktail tomatoes. Tomatoes are harvested weekly and graded on colour (red), shape (round), and firmness (solid). Fruit is packed into paper boxes and plastic baskets in 5, 7, 9, and 10 kg formats, with the 5 kg format being the most common, and is refrigerated prior to dispatch and sale.

12.8 Greenhouse Culture Cultivation Technology Summary

The annual production cycle runs from substrate buffering and planting in mid/late January, through automated fertigation, climate-controlled growth, biological pest control, and weekly harvesting, to crop clearance and greenhouse disinfection at the end of November – after which the cycle repeats.

12.9 Seed Production, Processing and Variety Maintenance

Not applicable in the strict sense of sowing-seed production: the company does not produce or process its own seed. Seeds and seedlings for both product lines are sourced externally from breeders in the Netherlands (Section 11).

13 Machinery Park

Harvesting and handling rely on harvesting trolleys, container carts, and lifting equipment. Crop support is provided by tomahooks with twine, truss support clips, and manual string winders. Climate control and fertigation are automated and integrated through Priva, a widely used Dutch greenhouse automation platform covering irrigation, fertigation, and climate control.

14 Introduction of Experiments

The company conducts variety and rootstock trials directly on-site through physical evaluation. At the time of the interview, five active trials were underway, including evaluation of a new rootstock, DR0141TX – a high-vigour rootstock developed by De Ruiter (Bayer) for high-tech, artificially lit glasshouse tomato production.

15 Work Management

15.1 Organizational Structure of the Farm

See Section 2 – the farm is organized into four operational divisions (each with a division manager and technician) and, at crop level, three departments: young plants (nursery), tomatoes, and paprika, under greenhouse production manager Ákos Magyar.

15.2 Personnel Conditions

See Section 3 for headcount and training practices. *[Not yet documented: breakdown of individual worker responsibilities within the tomato department – to be confirmed with the company.]*

15.3 Organizing Working Time

[Not yet documented in the interview: standard working hours, shift structure, and how rest days are organized, particularly during peak harvest periods. Recommended as a follow-up question with the company.]

15.4 Work Organization Solutions

[Not yet documented in the interview: specific work organization solutions (e.g. harvest crews, task rotation, quality checkpoints) used on the tomato line. Recommended as a follow-up question with the company.]

16 Economic Analysis

[Not yet documented – to be discussed only if the company is comfortable sharing this information. Open items include annual production volumes, major cost drivers, indicative selling prices, overall profitability, unit cost per kg, and efficiency indices. Present any figures obtained in a table in the Appendix.]

17 Marketing Directions of the Products

The company's primary buyer is Hódkertész Kft., a Hódmezővásárhely-based producer/sales organization, which in turn supplies major retail chains including Spar and Lidl. Tomatoes therefore reach consumers both through domestic supermarket chains and export markets, notably Spain, the Czech Republic, Germany, and Romania. The quality standard applied throughout the marketing chain requires fruit to be red, round, and solid.

18 Development Goals, Objectives and Plans

The company's main operational pressures at present are rising energy costs, input costs, and labour costs. *[Not yet documented: the company's specific plans or strategies for addressing these pressures, beyond the general 15-year phased development plan described in Section 1. Recommended as a follow-up question.]*

19 Use of Subsidy

The company has received government/EU agricultural subsidies covering approximately 50% of greenhouse construction costs. These funds have supported the construction of new greenhouses, the purchase of equipment, and the purchase of input materials. *[Not yet documented: the names of the specific subsidy programs or funding bodies involved, beyond the Növekedési Hitelprogram credit scheme referenced in Section 1 (which financed the company's founding, not its later construction subsidies). To be confirmed with the company.]*

20 Extension Services and Advocacy

The company works with agricultural advisors and extension services, and sells its produce through Hódkertész Kft. It is also reported to be a member of a TÉSZ (Termelői Értékesítő Szervezet – an EU-recognized Producer Organisation). *[Not yet documented: the full name of the specific TÉSZ organization(s) the company belongs to. To be confirmed with the company.]*

21 Developmental Aims and Plans

The company's development has followed a deliberate 15-year strategic plan carried out in stages since its founding in 2013, with a SWOT analysis conducted after each phase to guide subsequent investment decisions. Looking forward, the company's main challenges – rising energy, input, and labour costs – suggest that future investment will likely continue to prioritize automation and efficiency gains (such as the Priva climate/fertigation system already in place) over further area expansion, though this has not been confirmed directly with the company. *[Add your own concluding observations and suggestions here, based on your first-hand experience during the internship, as required by the guide.]*

References

[1] Szentesi Paradicsom Kft. *Company and Production Profile – Fűz-telep, Szegvár Greenhouse Facility*. On-site interview with company management, conducted during the engineering internship. *[Confirm the exact interview date for the citation.]* 2026.

Appendix A – Charts, Diagrams and Pictures

[Insert the organogram, greenhouse layout, production/economic charts, and photographs here. Example figure below – replace with your own image file.]

Appendix B – Internship Leader's Declaration

The Internship Leader's declaration about the completion of the Engineering Internship

The undersigned *[name of internship leader]* certify that *[student name]*, student of Faculty of Agriculture, University of Szeged, spent her/his Engineering Internship from *[start date]* until *[end date]* at *[name and headquarters of the company/firm]*.

I confirm that he/she accomplished their professional duties, wrote a daily diary about the tasks which corresponds to reality.

I grade his/her professional activity to be: *[excellent (5) / good (4) / satisfactory (3) / pass (2) / fail (1)]*

My professional opinion about the student's activities:

[...]

Date: *[...]* *[signature, stamp]*